QMST 4320.251
Project Report Paper

Exploring the Impact of Social Media on College Students' Mental Health

References:
https://www.kaggle.com/datasets/souvikahmed071/social-media-and-mental-health
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https://assets.cureus.com/uploads/review_article/pdf/31508/1612430233-1612430227-20210204-18204-83dotg.pdf
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Abstract

Over the last decade, social media has become an integral part of daily life for many, particularly among college students who strive to stay informed and connected. Although these platforms serve various purposes—from social updates to leisure activities—their excessive and aimless use has been linked to negative mental health outcomes. This paper explores the impact of social media on the mental health of college students, considering additional contributing factors. We use ordinal logistic models to analyze the relationships among these factors and mental health outcomes. Through our research, we aim to provide insights into how social media usage affects student well-being and discuss implications for mental health strategies in academic environments.

Introduction

Purpose of Report

Social media has become an integral part of modern life, serving as a crucial channel for receiving updates about global events, staying connected with friends and family, and establishing new connections across the globe. Despite its benefits, concerns about its negative repercussions persist.

Our goal is to investigate the impact of social media on mental health, with a particular focus on its association with depression. We plan to examine this issue by analyzing survey data that captures authentic, opinion-based perspectives. This research aims to shed light on the intricate effects of social media on our daily lives and psychological well-being.

In addition to examining the direct effects of social media on our emotions, we're also looking closely at another valuable source of data: a survey that focuses specifically on the relationship between university life and social media usage. This aspect is crucial, particularly because we're concentrating on university students as the central demographic of our study. Understanding how the academic environment and the unique challenges of university affect social media habits can provide us with important insights. This information will contribute to constructing a more comprehensive view of the role social media plays in a student's life and its possible influence on their mental health.

Research Methods

We used "R" to run an ordinal logistic regression on both sets of data. Then we proceed to analyze the t – values & p - values to determine the significant variables that affect depression. Furthermore, we analyzed the coefficients of those variables and evaluated how they impact depression.

Literature Review

Exploring the Influence of Social Media in Mental Health

In this study, we explore factors influencing mental health with a focus on social media engagement. We use secondary data that measures stress, depression, anxiety, sleep quality, and social support while controlling for demographics, lifestyle choices and social media usage. We use statistical models to analyze the relationships among these factors and mental health outcomes. Our research aims to offer insights into social media related interventions and strategies for enhancing mental well-being across diverse demographics.

Exploring the Influence of Social Media in Mental Health

The interconnection between social media usage and mental health outcomes has been a focal point of psychological and sociological research in recent years. This literature review seeks to distill the current findings, identifying both the adverse and beneficial effects of social media on mental well-being, and the various factors that modulate these effects.

Impact of Social Media on Mental Health

A significant body of research suggests a complex relationship between social media usage and mental health, with studies often yielding mixed results. On one hand, social media platforms can enhance social support, provide a sense of belonging, and facilitate social interaction, potentially buffering against mental health issues such as depression and anxiety. On the other hand, excessive social media use has been linked to negative outcomes, including increased stress, anxiety, depression, and poor sleep quality, particularly among adolescents and young adults. These findings underscore the dual-edged nature of social media, where its impact is deeply influenced by the manner and context of its use.

Factors Influencing the Social Media-Mental Health Relationship

Several key factors have been identified as moderating the relationship between social media engagement and mental health outcomes. These include individual differences (such as personality traits and pre-existing mental health conditions), the quality and nature of online interactions, and the amount of time spent on social media platforms. For instance, passive consumption of content and negative social comparisons on social media are associated with poorer mental health, while active engagement and positive interactions can enhance well-being.

Demographic and Lifestyle Variables

Demographic variables, including age, gender, and socioeconomic status, along with lifestyle choices such as physical activity and sleep habits, play a role in mediating the effects of social media on mental health. Younger individuals and those with certain lifestyle factors are more susceptible to the negative impacts of social media, highlighting the need for targeted interventions in these populations.

Methodological Considerations and Future Directions

The literature review also reveals methodological differences across studies, with varying measures of social media usage, mental health outcomes, and a wide range of statistical approaches. These differences underscore the complexity of quantifying social media's impact on mental health and suggest a need for more standardized methods of research in this area. Future studies should focus on longitudinal designs to better understand causal relationships and explore the efficacy of interventions aimed at reducing the negative effects of social media while promoting its positive aspects.

Conclusion

In conclusion, the current literature suggests a nuanced relationship between social media usage and mental health, influenced by a multitude of factors including the nature of online interactions, individual characteristics, and lifestyle choices. This review highlights the importance of considering these diverse factors in understanding and addressing the impacts of social media on mental well-being. It also points to the need for further research to elucidate these complex dynamics and develop effective strategies for mitigating the negative effects of social media, thereby harnessing its potential as a force for good in the mental health domain.

Data

Data Description

Social media and Mental Health dataset (Entries: 481 Variables: 21)

age: Respondent's age.
gender: Respondent's gender.
relationship_status: Respondent's relationship status.
occupation_status: Respondent's occupation status.
organization: Type of organizations the respondent is affiliated with.
socialmedia_use: Whether the respondent uses social media.
sm_platform: Social media platforms the respondent commonly uses.
sm_avgtime: The average time spent on social media every day.
sm_q1: How often the respondent uses social media without a specific purpose.
sm_q2: How often the respondent gets distracted by social media when busy.
sm_q3: Feelings of restlessness when not using social media for a while.
distracted_q: How easily the respondent is distracted (scale of 1 to 5).
bothered_q: How much the respondent is bothered by worries (scale of 1 to 5).
concentrate_q: Difficulty in concentrating on things.
selfcompare_q: How often the respondent compares themselves to other successful people through
social media (scale of 1 to 5). compare_q: Feelings about these comparisons generally speaking.

validation_q: How often the respondent seeks validation from social media.
depressed_q: Frequency of feeling depressed or down.
activites_q: Frequency of interest fluctuation in daily activities (scale of 1 to 5).
sleep_q: Frequency of facing issues regarding sleep (scale of 1 to 5).

Students' Mental Health Survey dataset (Entries: 7,022 Variables: 20)

Stress_Level: The level of stress experienced by the individual.
Depression_Score: The score representing the level of depression experienced by the individual.
Anxiety_Score: The score representing the level of anxiety experienced by the individual.
Sleep_Quality: The quality of sleep experienced by the individual .
Physical_Activity: The level of physical activity .
Diet_Quality: The quality of the individual's diet.
Social_Support: The level of social support received by the individual.
Substance_Use: The frequency of substance use such as alcohol, cigarettes or other drugs.
Family_History: Whether the individual has a family history of mental health issues.
Chronic_Illness
Financial_Stress: The level of financial stress experienced by the individual (between 0 and 5).
Semester_Credit_Load: The number of credits the individual is taking in the semester (15-30).

Descriptive Statistic

Social media and Mental Health dataset:

- Data includes several scale-based responses (1 to 5) about feelings, emotional fluctuations, and daily experiences related to mental health.

- Age:
    - Mean: 26.14
    - Standard Deviation: 9.92
    - Range: 13 to 91
- Scale-Based Responses:
    - Frequency of using social media without purpose:
        - Mean: 3.55
        - Standard Deviation: 1.10
        - Range: 1 to 5
    - Frequency of distraction by social media:
        - Mean: 3.32
        - Standard Deviation: 1.33
        - Range: 1 to 5

- Sleep issues:
    - Mean: 3.20
    - Standard Deviation: 1.46
    - Range: 1 to 5
- Fluctuation in interest in daily activities:
    - Mean: 3.17
    - Standard Deviation: 1.26
    - Range: 1 to 5

Social media and Mental Health dataset 2:

- Age ranges between 17 to 25 years with an average of approximately 21 years.
- CGPA on a scale of 0 to 4, with some missing values.
- Stress, depression, and anxiety scores are numerically rated, providing a quantifiable measure of mental health conditions.
- Financial stress and semester credit load are also quantified, allowing analysis of how these stressors correlate with mental health.
- CGPA:
    - Mean: 3.42
    - Standard Deviation: 0.58
    - Range: 1.78 to 4.00
- Stress Level:
    - Mean: 3.08
    - Standard Deviation: 1.31
    - Range: 1 to 5
- Depression Score:
    - Mean: 3.09
    - Standard Deviation: 1.21
    - Range: 1 to 5
- Anxiety Score:
    - Mean: 3.13
    - Standard Deviation: 1.23
    - Range: 1 to 5
- Financial Stress:
    - Mean: 3.26
    - Standard Deviation: 1.31
    - Range: 1 to 5
- Semester Credit Load:
    - Mean: 22.01
    - Standard Deviation: 4.36

- Range: 15 to 29

Data Analytics Application

In the context of our research report, the application of data analytics focuses primarily on understanding the complex relationship between social media use and its impact on mental health among college students. By employing ordinal logistic regression, our analysis methodologically breaks down various factors such as frequency of social media usage, its purpose, and accompanying mental health states like stress, depression, and anxiety. This analytical approach helps in identifying critical patterns and effects, providing a robust framework for interpreting how different dimensions of social media usage correlate with mental health outcomes.

This analysis is for formulating targeted interventions aimed at mitigating negative impacts while enhancing the positive aspects of social media use. The practical implementation of these data analytics techniques involved the use of the R programming language to handle and process the survey data effectively. By analyzing both t-values and p-values, we were able to determine the statistically significant variables that contribute to depression among students. This data analysis enables us to draw concrete conclusions about the role of social media in shaping mental health, particularly underlining the negative outcomes associated with aimless and excessive social media use. Through this analytical approach, our report contributes to a deeper understanding of the intricate dynamics between social media behavior and mental health, providing a data-driven basis for future policies and practices within academic environments.

Model Validation

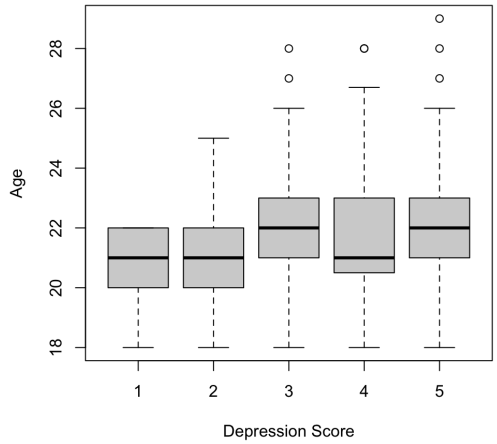
```
model <- polr(Depression_Score ~ Age + Course + Gender + CGPA + Stress_Level
              + Anxiety_Score + Sleep_Quality + Physical_Activity + Diet_Quality
              + Relationship_Status + Counseling_Service_Use  + Financial_Stress
              + Extracurricular_Involvement + Semester_Credit_Load , data = pdata2, Hess = TRUE)
```

	Value	Std. Error	t value	t_values	p_values
Age	0.0009892194	0.00546061	0.18115549	0.18115549	8.563734e-01
CourseComputer Science	1.3343149196	0.08715746	15.30924536	15.30924536	0.000000e+00
CourseEngineering	-0.0205369299	0.08503122	-0.24152223	-0.24152223	8.093229e-01
CourseLaw	-0.0024634865	0.08295358	-0.02969717	-0.02969717	9.763292e-01
CourseMedical	-0.0563085658	0.07790413	-0.72279310	-0.72279310	4.703956e-01
CourseOthers	-0.0385563793	0.09383511	-0.41089503	-0.41089503	6.814561e-01
GenderMale	-0.0129785288	0.04225153	-0.30717299	-0.30717299	7.589345e-01
CGPA	-0.1672818645	0.07363661	-2.27172142	-2.27172142	2.384353e-02
Stress_Level	-0.0157592837	0.01358672	-1.15990368	-1.15990368	2.470520e-01
Anxiety_Score	-0.0061373656	0.01353766	-0.45335511	-0.45335511	6.506356e-01
Sleep_QualityGood	-0.0259221026	0.04490940	-0.57720883	-0.57720883	5.642509e-01
Sleep_QualityPoor	-0.0284539372	0.07563177	-0.37621674	-0.37621674	7.070337e-01
Physical_ActivityLow	-0.0336301211	0.06121093	-0.54941364	-0.54941364	5.831489e-01
Physical_ActivityModerate	0.0189566508	0.05612241	0.33777328	0.33777328	7.357810e-01
Diet_QualityGood	0.1296248115	0.05465193	2.37182490	2.37182490	1.836022e-02
Diet_QualityPoor	0.0368372935	0.05497047	0.67012881	0.67012881	5.033145e-01
Relationship_StatusMarried	0.0596791493	0.06131939	0.97325082	0.97325082	3.312482e-01
Relationship_StatusSingle	0.0163125129	0.04881498	0.33417020	0.33417020	7.384952e-01
Counseling_Service_UseNever	-0.0729416953	0.07333357	-0.99465632	-0.99465632	3.207413e-01
Counseling_Service_UseOccasionally	-0.0767511786	0.07864816	-0.97588010	-0.97588010	3.299457e-01
Financial_Stress	-0.0176731093	0.01232811	-1.43356216	-1.43356216	1.527858e-01
Extracurricular_InvolvementLow	0.0425130634	0.06036138	0.70430906	0.70430906	4.818114e-01
Extracurricular_InvolvementModerate	0.0592967874	0.05578064	1.06303535	1.06303535	2.886595e-01
Semester_Credit_Load	0.0106034537	0.00485148	2.18561218	2.18561218	2.965214e-02
0|1	-1.8035847255	0.33228376	-5.42784489	-5.42784489	1.215011e-07
1|2	-0.8158572534	0.33168520	-2.45973365	-2.45973365	1.449368e-02
2|3	-0.0240043869	0.33160218	-0.07238911	-0.07238911	9.423427e-01
3|4	0.9326007771	0.33185279	2.81028461	2.81028461	5.290220e-03
4|5	1.8557798351	0.33271693	5.57765382	5.57765382	5.632675e-08

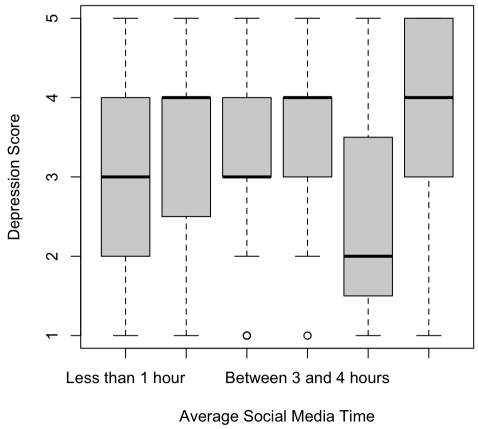
```
model <- polr(depressed_q ~ age + gender + relationship_status + sm_avgtime + sm_q1 + sm_q2
              + sm_q3 + validation_q + activites_q + sleep_q + selfcompare_q + distracted_q
              + bothered_q + concentrate_q , data = pdata, Hess=TRUE)
```

	Value	Std. Error	t value	t_values	p_values
age	0.08848827	0.05611879	1.5768030	1.5768030	1.159199e-01
genderMale	-0.08668386	0.22872616	-0.3789853	-0.3789853	7.049733e-01
relationship_statusMarried	-0.67858789	0.50982290	-1.3310267	-1.3310267	1.842150e-01
relationship_statusSingle	-0.09826846	0.24784753	-0.3964875	-0.3964875	6.920340e-01
sm_avgtimeBetween 2 and 3 hours	0.46528849	0.44263905	1.0511691	1.0511691	2.940471e-01
sm_avgtimeBetween 3 and 4 hours	-0.10817736	0.43224668	-0.2502676	-0.2502676	8.025558e-01
sm_avgtimeBetween 4 and 5 hours	0.28585506	0.46644312	0.6128401	0.6128401	5.404574e-01
sm_avgtimeLess than an Hour	-0.39273235	0.69199870	-0.5675334	-0.5675334	5.707865e-01
sm_avgtimeMore than 5 hours	0.56334396	0.43502577	1.2949669	1.2949669	1.963511e-01
sm_q1	0.08536752	0.12577117	0.6787527	0.6787527	4.978304e-01
sm_q2	0.12631845	0.11964563	1.0557716	1.0557716	2.919423e-01
sm_q3	-0.05892786	0.10506711	-0.5608593	-0.5608593	5.753221e-01
validation_q	0.06067634	0.09887389	0.6136740	0.6136740	5.399069e-01
activites_q	0.32026150	0.11023006	2.9053916	2.9053916	3.947443e-03
sleep_q	0.12734106	0.07916574	1.6085374	1.6085374	1.087945e-01
selfcompare_q	0.26688173	0.09042946	2.9512699	2.9512699	3.420392e-03
distracted_q	-0.04508191	0.13909769	-0.3241025	-0.3241025	7.460917e-01
bothered_q	0.52366014	0.11224697	4.6652495	4.6652495	4.693304e-06
concentrate_q	0.18829803	0.12952925	1.4537104	1.4537104	1.470969e-01
1|2	4.17554658	1.42790304	2.9242508	2.9242508	3.722407e-03
2|3	5.77176132	1.44583562	3.9919900	3.9919900	8.288276e-05
3|4	7.48386277	1.47140851	5.0861897	5.0861897	6.532300e-07
4|5	9.03827080	1.48920769	6.0691809	6.0691809	3.969925e-09

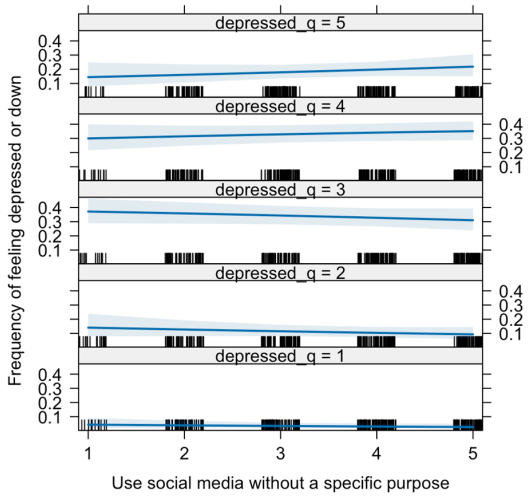
Age vs. Depression Score

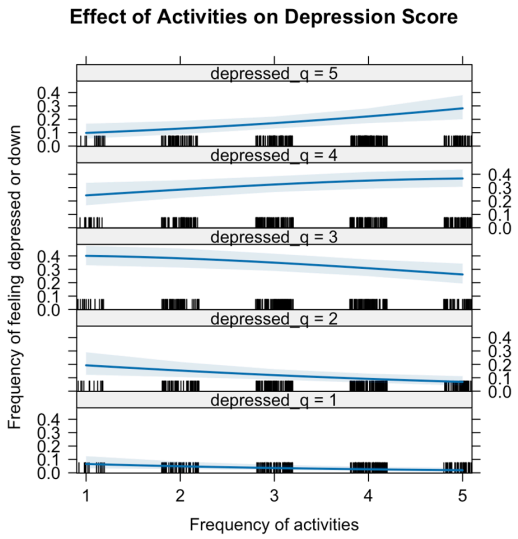


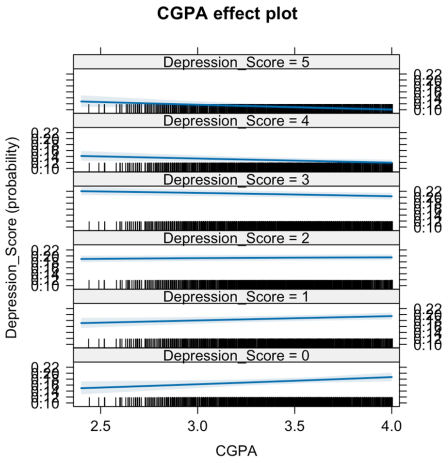
Average Social Media Time vs. Depression Score

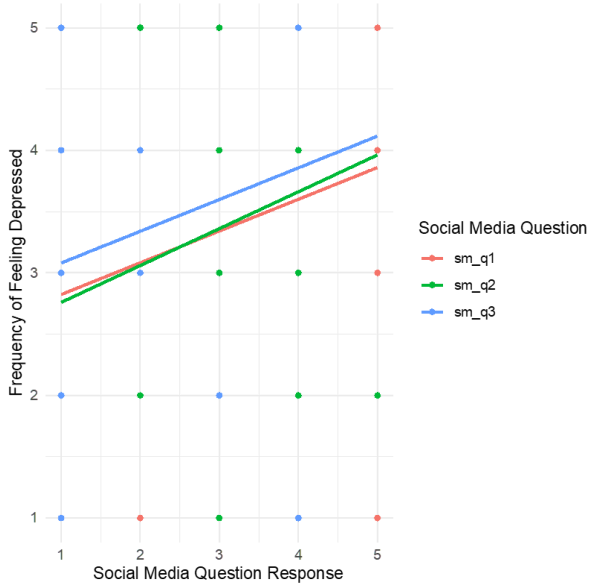


Effect of Social Media Use on Depression Score




Effect of Activities on Depression Score
depressed_q = 5
depressed_q = 4
depressed_q = 3
depressed_q = 2
depressed_q = 1
Frequency of feeling depressed or down
Frequency of activities


CGPA effect plot
Depression_Score = 5
Depression_Score = 4
Depression_Score = 3
Depression_Score = 2
Depression_Score = 1
Depression_Score = 0
Depression_Score (probability)
CGPA


Frequency of Feeling Depressed
Social Media Question Response
Social Media Question
sm_q1
sm_q2
sm_q3

Discussion & Conclusion

Insights:

•With more time spent on social media concludes a higher depression score among the survey participants.
•With a greater use of social media without a purpose affects depression.
•More participants record a higher depression score with a higher score of use of social media without a purpose.
•Similarly, more participants who compare themselves & seek validation from social media also result to a higher depression scores.
•Our secondary data shows that with a higher GPA there is a lower probability of depression.

Conclusion:

Our research shows that social media has become a kind of escape for many people, a place they go to avoid real-life problems. However, our findings also show that spending more time on social media can make you feel more depressed. Even when we looked at other things that could affect how you feel, like stress from school or where you come from, social media still had a big impact on mental health. So, it's clear from our study that social media really does affect how we feel inside.